

Biology Principles - Chapter 8: Energy and Life

Photosynthesis is the fundamental biological process by which autotrophic organisms convert light energy into chemical energy stored in glucose molecules. In green plants, this conversion takes place primarily inside specialized organelles called chloroplasts.

The thylakoid membranes contain chlorophyll pigments that absorb blue and red wavelengths while reflecting green light. During the light-dependent reactions, water molecules are split through photolysis, releasing molecular oxygen as a vital byproduct.

Cellular respiration operates conversely by breaking down carbohydrates in mitochondria to yield adenosine triphosphate (ATP) for cellular work.

The Calvin cycle, also known as the light-independent reactions, occurs within the stroma of chloroplasts. Carbon dioxide is captured through carbon fixation catalyzed by the enzyme RuBisCO, ultimately producing glyceraldehyde 3-phosphate (G3P).

Environmental factors such as ambient temperature, carbon dioxide concentration, and light irradiance substantially influence photosynthetic efficiency and overall agricultural crop yields.